

Exploring the Effects of Fantasy Level of Avatars on User Perception and Behavior

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ABSTRACT

Embodying avatars in virtual reality (VR) has transformed human experiences, such as in medicine and education. However, there is limited information about users' self-identifications and perceptions of highly fantastical avatars. This pilot study explored the impact of avatar types of low and high fantasy levels on adults' perceptions and behaviors. Participants ($N = 18$) engaged in a VR experience with either a human or blue Muppet avatar to complete body movement tasks, a cube-touching game, and free-form exploration. Findings showed that participants in the high fantasy avatar condition reported higher identification with their avatar and more interest in social-emotional activities relative to the low fantasy human avatar condition. Across both conditions, participants stood extremely close to the virtual mirror. Additionally, we report on participants preferences and priorities for their future avatars. This offers insights for future research on avatar design, with implications for more engaging VR experience.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**;

KEYWORDS

Avatar, virtual reality, human-centered design, user experience

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1 INTRODUCTION

Virtual reality (VR) avatars have revolutionized digital interaction, demonstrating the potential to enhance a variety of domains such as education, therapy, collaborative learning, and entertainment [8]. Within VR, the specific type of avatars that users embody can impact their emotions [2, 10] self- and social perceptions [3], and behaviors [6, 9, 14] in different ways. For example, embodying

realistic full-body avatars significantly improved people's memory tasks compared to no avatars [7] while there was another research showed that a faithful reproduction of the participant's hand solicited lower body ownership over the avatar than using a cartoonish avatar [5]. Besides studies around the embodiment of realistic avatars, fantasy in VR typically is defined as experiences that go beyond realism. The unique affordances of VR allow users to embody avatars that represent different levels of fantasy such as embodying a human (e.g., low fantasy level) or controlling the body of a dragon (e.g., high fantasy level). Some evidence shows that controlling an avatar that is impossible can impact users' perceptions. For example, a participant embodying superheroes with the ability to fly performed better in altruistic behaviors during the VR experience as well as after the experience ended [8]. However, there remains gaps on users' self and social-emotional perceptions, preferences, and behaviors about avatars that represent high levels of fantasy compared to low levels of fantasy. Meta and Apple released their humanoid avatars for wide business usage. With the continued development of VR for experiences beyond gaming, it is important to understand how various avatar types can positively impact user's perceptions and behaviors. This will be beneficial in promoting the advancement of user experience in the VR space.

2 DESCRIPTION

To test and gain further insights into avatar type on user's experiences, a pilot study using a mixed methods and between-participants design was conducted. The study explored how the type of avatar embodiment (i.e., low fantasy and high fantasy) in VR impacted users' perceptions (i.e., self-presence, social-emotional perceptions, avatar design preferences), and behaviors (i.e., walking towards the virtual mirror). Participants were randomly assigned to either a low fantasy realistic humanoid avatar (i.e., toon teen), or a highly fantastical avatar (i.e., a blue Muppet) and completed a variety of tasks during a 20- to 30-minute VR experience.

2.1 Equipment and Virtual Environment

The VR experience was built and programmed within Unity 3D which utilized Meta Quest 2 head-mounted display (HMD) with two hand controllers. Two avatar types were created for the experience: (a) human (i.e., animated humanoid character) which was designed with Toon Teen package from Unity Store and (b) non-human (i.e., blue Muppet, GroverTM) which was provided by Sesame Workshop. Human avatars were designed in skin tone, hair style and hair color to fit a diverse range of people characteristics. Both avatar types were aligned with the same height and body movement animations to ensure consistency.

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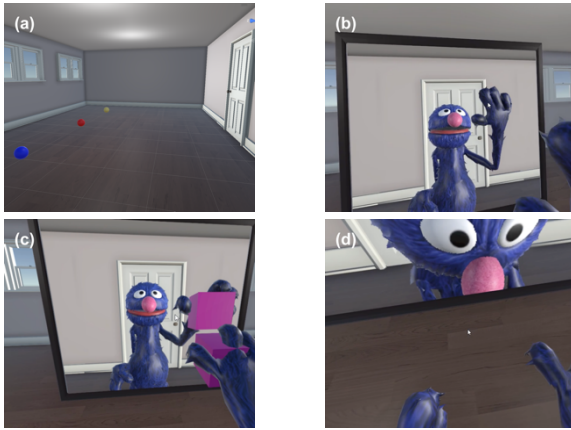


Figure 1: Four phases of the VR experience embodying the non-human avatar: (a) orientation, (b) avatar warm-up, (c) go/no-go task, (d) free exploration where the participant stood extremely close to the virtual mirror.

2.2 Participants

We recruited 18 participants from the University of Texas at Austin and its nearby surrounding area. They identified and reported their age ($M = 24.78$ years, $SD = 3.35$ years, range = 18-33 years of age), gender identity (12 female, 5 male, 1 non-binary), race/ethnicity, and their VR and gaming experience within the past 6-months.

2.3 Procedure

All participants provided informed consent, and confirmed they had no health conditions including epilepsy and disorientation that could lead to discomfort during the VR testing experience. First, participants completed a pre-test questionnaire measuring demographics and simulator sickness. Next, they completed a four-phase VR experience (Figure 1): (a) a VR orientation, identifying three different colored spheres (they could not see their avatar's body at this time), (b) an avatar warm-up, where they did body movements as instructed by the researcher and saw their avatar's body from a first-person point of view in front of a virtual mirror, (c) a go/no-go task, where they touched virtual colored cubes, following specific instructions [12], and (d) a 1-minute of free time to explore the environment. After completing the go/no-go task, participants responded to self-presence questions asked by the researcher, measuring how closely they identified with their embodied avatar [4, 11, 13]. Response options ranged from not at all (1) to very strongly (5). To assess participant self-identification to their avatar, the closest distance (i.e., minimum distance) their avatar stood to the virtual mirror was passively collected through the VR system's head tracking. Finally, after the VR experience, participants answered scaled questions for social-emotional use cases for their avatar (e.g., interest in using avatar to manage emotions) and their preferences for future avatars on realism and bodies [12, 13]. All sessions were video recorded for analysis, with all procedures approved by the university's Institutional Review Board. Participants received a \$20 gift card for their involvement.

Table 1: Means and standard deviations of quantitative measures.

Measures (SD)	Human (N=9)	Non-human (N=9)	All (N=18)
SP	2.94 (0.35)	3.61 (0.77)	3.28 (0.67)
Mirror	-0.14 (0.39)	-0.17 (0.88)	-0.16 (0.66)
SE 1	2.78 (0.97)	3.44 (1.13)	3.11 (1.08)
SE 2	2.22 (1.30)	2.78 (1.39)	2.33 (1.34)
SE 3	1.67 (0.71)	2.56 (1.51)	2.11 (1.23)
SE 4	2.89 (1.27)	4.11 (0.93)	3.50 (1.25)

3 RESULT

We calculated the mean and standard deviations of each category to explore the effect of avatar type on perception. Table 1 includes self-presence (SP) (e.g., how much they felt they were the avatars), mean minimum distance to the virtual mirror (Mirror), and interest in various social-emotional activities, including their desire to play games with friends as avatar (SE 1), feel better when sad as avatar (SE 2), share feelings with a friend as avatar (SE 3), talk to a stranger as avatar (SE 4). All measured on a scale from 1 to 5, except for Mirror which was measured in meters.

Additionally, qualitative thematic coded analysis [1] revealed that the avatar's face, specifically facial expressions, and facial characteristics, as the most important physical attribute to participants ($N = 12$), with having legs and leg movement as the most important feature ($N = 6$). In their descriptions, participants tended to focus on either the upper body (face, hands, arms, and hair) or lower body (legs), with few describing wanting both ($N = 5$).

4 CONCLUSION AND FUTURE WORK

Overall, trends showed that on average adults embodying the non-human avatar reported higher levels of self-presence and greater interest in using the avatar for all the cases for social-emotional interactions relative to the human avatar condition showing great potential in high-fantasy avatar design in social games and representations. Additionally, when given the opportunity, users express interest in closely exploring their digital body, regardless of avatar type. Future research could collect more detailed information on adults' preferences regarding avatar customization and specifications. Further data collection and analysis will provide clearer insights on the impact of realistic versus fantastical avatars on users' self-, social perceptions and behaviors in the virtual space.

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